Second-Order and Quasi-Newton Methods

Optimization Methods, Week 5

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Learning objectives

After this lecture, you should be able to

- derive Newton's step from a quadratic model;
- explain local quadratic convergence and its assumptions;
- repair unsafe Newton steps using damping and regularization;
- construct DFP and BFGS updates from secant information;
- explain how L-BFGS, L-BFGS-B, and Hessian-vector products reduce cost.

Second-order Taylor model

For twice differentiable $f : \mathbb{R}^n \to \mathbb{R}$, near x,

$$f(x+p) = f(x) + \nabla f(x)^\top p + \frac{1}{2} p^\top \nabla^2 f(x) p + o(\|p\|^2).$$

Write $g = \nabla f(x)$ and $H = \nabla^2 f(x)$. The local quadratic model is

$$m_x(p) = f(x) + g^\top p + \frac{1}{2} p^\top H p.$$

The gradient controls local slope. The Hessian controls how slope changes and sets the local geometry.

Newton's step

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Geometric meaning

Gradient descent uses spherical local geometry:

$$p_{GD} = -\alpha g.$$

Newton's method uses the Hessian metric:

$$p_N = -H^{-1} g.$$

When $H \succ 0$, directions with high curvature receive smaller components. For a positive definite quadratic, Newton rescales every eigendirection exactly and reaches the minimizer in one iteration.

One-step convergence on a quadratic

Theorem

For $f(x) = \frac{1}{2}x^\top Qx - b^\top x + c$ with $Q \succ 0$, one full Newton step reaches $x^* = Q^{-1}b$ from any x.

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Local quadratic convergence

Theorem

Suppose x^* is a stationary point, $\nabla^2 f(x^*) \succ 0$, and the Hessian is locally M-Lipschitz. In a sufficiently small neighborhood, full Newton steps satisfy

$$\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2.$$

Quadratic convergence means that the number of correct digits roughly doubles at each iteration once the iterates enter the local region.

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Why Newton can fail

The Newton direction satisfies

$$g^\top p_N = -g^\top H^{-1} g.$$

This quantity is negative when $H \succ 0$, but it can be positive when H is indefinite. Further problems include

- a singular or nearly singular Hessian;
- attraction to a saddle point or maximum;
- a good local model paired with a step that leaves the local region;
- numerical error in an ill-conditioned linear solve.

An indefinite Hessian can point uphill

Consider $f(x, y) = \frac{1}{2}x^2 - y^2 + \frac{1}{4}y^4$. At $(0, 0.5)$,

$$g = (0, -0.875)^\top, \qquad H = \operatorname{diag}(1, -1.25).$$

The Newton direction is $p_N = (0, -0.7)^\top$, and

$$g^\top p_N = 0.6125 > 0.$$

It is an ascent direction. A line search that assumes descent cannot repair the direction; the model itself needs modification.

Damped Newton with Armijo line search

Given a descent direction p_k, choose $\alpha_k = 1, \beta, \beta^2, \ldots$ until

$$f(x_k + \alpha_k p_k) \leq f(x_k) + c\alpha_k \nabla f(x_k)^\top p_k.$$

Then set $x_{k+1} = x_k + \alpha_k p_k$. Far from the solution, damping protects global progress. Near a regular minimizer, the full step $\alpha_k = 1$ is eventually accepted, preserving quadratic convergence.

Hessian regularization

Replace the Newton equation with

$$(H + \lambda I)p = -g, qquad \lambda \geq 0.$$

Choose λ large enough that $H + \lambda I \succ 0$. Then

$$g^\top p = -g^\top (H + \lambda I)^{-1} g < 0.$$

As $\lambda \to 0$, the step approaches Newton. As $\lambda \to \infty$, $p \approx -\lambda^{-1} g$, a short gradient step.

Levenberg-Marquardt for nonlinear least squares

For $f(x) = \frac{1}{2}\|r(x)\|^2$, with Jacobian $J(x)$,

$$\nabla f(x) = J^\top r, qquad \nabla^2 f(x) = J^\top J + \sum_i r_i \nabla^2 r_i.$$

Gauss-Newton drops the second term. Levenberg-Marquardt solves

$$(J^\top J + \lambda I)p = -J^\top r.$$

The regularization stabilizes rank-deficient Jacobians and interpolates between Gauss-Newton and gradient descent.

Dense second-order cost

For n parameters:

Operation	Dense cost	Memory
form Hessian	problem dependent, often $O(n^2)$ derivatives	$O(n^2)$
factorize Hessian	$O(n^3)$	$O(n^2)$
solve after factorization	$O(n^2)$	$O(n^2)$
Hessian-vector product	often comparable to a few gradients	$O(n)$ extra

Explicit Newton is attractive for moderate dimension and expensive objectives, but infeasible for models with millions of parameters.

Quasi-Newton idea

Quasi-Newton methods build curvature estimates from gradient differences. Define

$$s_k = x_{k+1} - x_k, qquady_k = \nabla f(x_{k+1}) - \nabla f(x_k).$$

The fundamental theorem of calculus gives

$$y_k = \left(\int_0^1 \nabla^2 f(x_k + ts_k) dt \right) s_k.$$

Thus y_k is the action of an average Hessian on s_k.

Secant conditions

If B_k approximates the Hessian and G_k approximates its inverse, require

$$B_{k+1}s_k = y_k, qquad G_{k+1}y_k = s_k.$$

One vector equation cannot determine an entire matrix. DFP and BFGS add symmetry, low-rank change, and positive-definiteness preservation to select a unique practical update.

BFGS updates

The Hessian form is

$$B_{k+1} = B_k - \frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}.$$

The inverse form used for directions is

$$G_{k+1} = (I - \rho_k s_k y_k^\top) G_k (I - \rho_k y_k s_k^\top) + \rho_k s_k s_k^\top.$$

Then $p_k = -G_k \nabla f(x_k)$, usually combined with a line search.

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For nonzero z, write $V = I - \rho s y^\top$ and $z^\top G_{k+1} z = (V^\top z)^\top G_k (V^\top z) + \rho (s^\top z)^2$.

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Curvature condition and line search

The condition $y_k^\top s_k > 0$ preserves positive definiteness. It holds automatically for strongly convex functions. For general smooth objectives, a line search enforcing the Wolfe curvature condition gives

$$\nabla f(x_k + \alpha_k p_k)^\top p_k \geq c_2 \nabla f(x_k)^\top p_k, qquad c_2 < 1,$$

which implies $y_k^\top s_k > 0$ for a descent direction. Practical implementations skip or damp updates when curvature is too small.

L-BFGS and limited memory

Full BFGS stores an $n \times n$ matrix. L-BFGS stores only the latest m pairs (s_i, y_i), usually m between 5 and 20. The two-loop recursion computes

$$p_k = -G_k g_k$$

without forming G_k, in $O(mn)$ time and memory. Old pairs are discarded, so the method trades some curvature information for scalability.

L-BFGS two-loop recursion

Set $q = g_k$. For stored pairs from newest to oldest:

$$\alpha_i = \rho_i s_i^\top q, qquad q \leftarrow q - \alpha_i y_i.$$

Apply an initial scaling $r = \gamma_k q$, commonly $\gamma_k = (s_{k-1}^\top y_{k-1})/(y_{k-1}^\top y_{k-1})$. Then scan oldest to newest:

$$\beta_i = \rho_i y_i^\top r, qquad r \leftarrow r + s_i(\alpha_i - \beta_i).$$

Return $p_k = -r$.

L-BFGS-B for box constraints

For bounds $\ell \leq x \leq u$, stationarity uses the projected gradient. L-BFGS-B combines limited-memory curvature with

- projection onto the box;
- a generalized Cauchy point along the projected search path;
- a subspace minimization over variables not fixed at a bound;
- a bound-aware line search.

Clipping an unconstrained L-BFGS step does not reproduce this algorithm.

Second-order methods in machine learning

They are useful when gradients are expensive, high accuracy matters, and parameter dimension is moderate. Examples include logistic regression, generalized linear models, scientific inverse problems, and final refinement after first-order training.
They are less attractive for huge stochastic neural networks: Hessians are large and indefinite, mini-batch curvature is noisy, and low-accuracy solutions may suffice.
Matrix-free products, Gauss-Newton structure, and L-BFGS can still help in selected regimes.